
Current Position

- Aug. 2022 – **NeuroAI Scholar**, Cold Spring Harbor Laboratory, Long Island, NY
Present *Independent postdoctoral scholar*
Bridging computer science and neuroscience in artificial neural networks to build more energy-efficient computing. Collaborating with neuroscientists to study facial expressions in mice.

Education

- 2016 – 2022 **Ph.D. in Electrical Engineering**, University of Wisconsin-Madison,
Dissertation: "Building Energy Efficient Computers with Brain-Inspired Computing Models"
- 2016 – 2019 **M.S. in Electrical Engineering**, University of Wisconsin-Madison,
- 2012 – 2016 **B.S. in Computer Engineering and Continuous Applied Mathematics**,
Rose-Hulman Institute of Technology,
Magna Cum Laude

Academic Experience

Research Experience

- 2016 – 2022 **Research Assistant**, University of Wisconsin-Madison,
Advised by: Dr. Mikko Lipasti
Worked on energy-efficient computing paradigms in stochastic computing and neural networks.
- 2014 – 2016 **Research Assistant**, Rose-Hulman Institute of Technology,
Advised by: Dr. Mario Simoni and Dr. Daniel Chang
Developed configurable architecture for simulating Hodgkin-Huxley neural systems.

Teaching Experience

- Sep. 2021 – **Lecturer**, University of Wisconsin-Madison,
Dec. 2021 ECE252: Introduction to Computer Engineering
Instructed a course section on introductory material for computer engineering.
- Sep. 2018 – **Teaching Assistant**, University of Wisconsin-Madison,
Aug. 2019 ECE532: Matrix Methods for ML
Taught as an in-class TA for a flipped class on linear algebra and machine learning.
- Sep. 2017 – **Teaching Assistant**, University of Wisconsin-Madison,
Dec. 2018 ECE315: Intro. to Microprocessor Laboratory
Instructed lab course on designing, assembling, and programming a printed circuit board.
- Jan. 2017 – **Teaching Assistant**, University of Wisconsin-Madison,
May 2018 ECE353: Intro. to Microprocessor Systems
Taught as an in-class TA for a flipped class on embedded systems.
- Mar. 2016 – **Teaching Assistant**, Rose-Hulman Institute of Technology,
May 2016 ECE530: Advanced Microcomputers
- Mar. 2014 – **Teaching Assistant**, Rose-Hulman Institute of Technology,
May 2014 CSSE332: Operating Systems

Manuscripts

Preprints (in-progress work)

- May 2025 **Walking the Weight Manifold: a Topological Approach to Conditioning Inspired by Neuromodulation**, *Preprint*,
Authors: A. S. Benjamin*, K. Daruwalla*, C. Pehle*, A. Zador
- Nov. 2021 **Accelerating Deep Learning with Dynamic Data Pruning**, *Preprint*,
Authors: R. S. Raju, K. Daruwalla, M. Lipasti

Publications

- Apr. 2026 **Cheese3D: Sensitive Detection and Analysis of Whole-Face Movement in Mice**, *Nature Neuroscience*,
Authors: K. Daruwalla*, I. N. Martin*, L. Zhang, D. Naglič, A. Frankel, C. Rasgaitis, R. Zhao, X. Zhang, Z. Ahmad, J. C. Borniger, X. H. Hou
- Dec. 2024 **Continual learning with the neural tangent ensemble**, *NeurIPS 2024* (🔦 spotlight paper),
Authors: A. S. Benjamin, C. Pehle, K. Daruwalla
- Dec. 2024 **Delays in generalization match delayed changes in representational geometry**, *NeurIPS 2024 Workshop on Unifying Representations in Neural Models (UniReps)*,
Authors: X. Zheng, K. Daruwalla, A. S. Benjamin, D. Klindt
- May 2024 **Information Bottleneck-Based Hebbian Learning Rule Naturally Ties Working Memory and Synaptic Updates**, *Frontiers in Computational Neuroscience*,
Authors: K. Daruwalla, M. Lipasti
- Apr. 2024 **BitFit: Bitstream-Aware Training for Stochastic Neural Networks**, *Second Workshop on Unary Computing (WUC 24)*,
Authors: N. Joshi, K. Daruwalla, M. Lipasti
- Feb. 2023 **Energy-Efficient Bayesian Inference Using Bitstream Computing**, *IEEE Computer Architecture Letters*,
Authors: S. Khoram, K. Daruwalla, M. Lipasti
- Nov. 2019 **BitSAD v2: Compiler Optimization and Analysis for Bitstream Computing**, *ACM Transactions on Architecture and Code Optimization (TACO)*,
Authors: K. Daruwalla, H. Zhuo, R. Shukla, M. Lipasti
- Jun. 2019 **BitBench: A Benchmark for Bitstream Computing**, *Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES 19)*,
Authors: K. Daruwalla, H. Zhuo, C. Schulz, M. Lipasti
- Jun. 2019 **Resource Efficient Navigation Using Bitstream Computing**, *First ISCA Workshop on Unary Computing (WUC 19)*,
Authors: K. Daruwalla, H. Zhuo, M. Lipasti
- Jun. 2019 **BitSAD: A Domain-Specific Language for Bitstream Computing**, *First ISCA Workshop on Unary Computing (WUC 19)*,
Authors: K. Daruwalla, H. Zhuo, M. Lipasti
- Jan. 2019 **A quantitative analysis of the performance of computing architectures used in neural simulations**, *Journal of Neuroscience Methods*,
Authors: K. Daruwalla, N. Olivero, A. Pluger, S. Rao, D. W. Chang, M. Simoni

Presentations

Invited and contributed talks

- Jun. 2023 **Intro. to FluxML and Machine Learning in Julia** [Invited]
Data Umbrella Seminar Series, Online (YouTube Live)
- Feb. 2022 **Building Energy-Efficient Computers** [Invited]
Cold Spring Harbor Lab NeuroAI Seminar, Long Island, NY
- Jan. 2020 **BitSAD v2: A Domain-Specific Language for Bitstream Computing** [Conference]
High-performance Embedded Architecture and Compilation Conference (HiPEAC 20), Bologna, Italy
- Oct. 2019 **BitSAD v2**
Industry Affiliates Meeting, Madison, WI
- Jun. 2019 **Resource Efficient Navigation Using Bitstream Computing** [Workshop]
First ISCA Workshop on Unary Computing (WUC 19), Phoenix, AZ
- Jun. 2019 **BitBench: A Benchmark for Bitstream Computing** [Conference]
Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES 19), Phoenix, AZ
- Oct. 2018 **BitSAD: A Domain-specific language for Bitstream Computing**
Industry Affiliates Meeting, Madison, WI
- Oct. 2017 **Seeing Through the FoG: A Biologically Inspired Navigation System**
Industry Affiliates Meeting, Madison, WI

Posters

- Jul. 2025 **Neuromodulation implies a manifold of model weights** [Conference]
NeuroAI in Seattle 2025, Seattle, Washington
- Mar. 2025 **Cheese3D: sensitive detection and analysis of whole-face movement in mice** [Conference]
Computational and Systems Neuroscience (COSYNE 2025), Montreal, QC
- Sep. 2024 **Generative modeling of trained networks as an analogy for neuronal development** [Conference]
From Neuroscience to Artificial Intelligence (NAISys 2024), Long Island, NY
- Mar. 2024 **The dynamics of interpretable 3D facial features reflect hidden neural and physiological states in mice** [Conference]
Neuronal Circuits, Long Island, NY
- Nov. 2021 **A Biologically Plausible Learning Rule Based on the Information Bottleneck** [Workshop]
Spiking Neural networks as Universal Function Approximators (SNUFA 21), Virtual
- Oct. 2019 **BitSAD v2**
Industry Affiliates Meeting, Madison, WI
- Jun. 2019 **BitBench: A Benchmark for Bitstream Computing** [Conference]
Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES 19), Phoenix, AZ
- Oct. 2018 **BitSAD: A Domain-specific language for Bitstream Computing**
Industry Affiliates Meeting, Madison, WI
- Oct. 2017 **Seeing Through the FoG: A Biologically Inspired Navigation System**
Industry Affiliates Meeting, Madison, WI

Nov. 2016 **Drone Control with Map-Seeking Circuits**

Industry Affiliates Meeting, Madison, WI

General public talks

Jun. 2024 **Cocktails & Chromosomes: Kyle Daruwalla**

Cocktails & Chromosomes Series, Long Island, NY

Apr. 2024 **Research & AI Panel Discussion**

[Panel]

Long Island Artificial Intelligence Conference, Long Island, NY

Jun. 2023 **Cocktails & Chromosomes: Kyle Daruwalla**

Cocktails & Chromosomes Series, Long Island, NY

Jan. 2023 **Computers, Brains, and the In-Between**

Cold Spring Harbor Lab DeMystifying Science Series, Long Island, NY

Awards

Sep. 2021 Second place – AFRL xView2 Overhead Imagery ML Hackathon

May 2018 Gerald Holdridge Teaching Excellence Award

Feb. 2016 Honor Student Award

Service, Mentorship, and Leadership

Mentorship

Research mentor • Catherine Rasgaitis (undergraduate) • Xingyu Zhang (graduate student)
• Irene Nozal Martin (graduate student)

Open-source mentor (*as part of Google Summer of Code for FluxML*)

• Abhirath Anand • Chandra Kiran
• Jiangeng • Saksham Rastogi

Service and Leadership

Program Committee Second Workshop on Unary Computing, Third Workshop on Unary Computing

Reviewer NeurIPS, UniReps, ICLR, Workshop on Unary Computing

2021 – 2022 Graduate student representative on departmental committee

2019 – 2020 Vice-President of ECE Graduate Student Association

2017 – 2019 President of ECE Graduate Student Association

2016 – 2017 Public Relations Officer of ECE Graduate Student Association

2012 – 2016 President of Linux Users Group

2014 – 2016 Operations Manager of Rose Tech Radio Club

2015 – 2016 Founder of Rose Maker Lab

Press

Jun. 2024 **Can AI learn like us?**, *CSHL Press*, 20 June 2024. Available: <https://www.cshl.edu/can-ai-learn-like-us/>

Jul. 2024 **Advancing AI's boundaries at Cold Spring Harbor Labs**, *Long Island Herald*, 5 July 2024. Available: <https://www.liherald.com/stories/advancing-ais-boundaries-at-cold-spring-harbor-labs,208824>

Industry Experience

- Jun. 2020 – **Research Co-op Intern, AMD, Austin, TX**
- Dec. 2020 Explored neural network sensitivity to input perturbations in the context of scientific computing.
- May 2017 – **Digital Design Intern, Texas Instruments, Dallas, TX**
- Aug. 2017 Performed IP design for multimedia IPs.
- Jun. 2016 – **Digital Design Intern, Texas Instruments, Dallas, TX**
- Aug. 2016 Performed design verification for high speed I/O IPs.
- Sep. 2015 – **Project Intern, Rose-Hulman Ventures, Terre Haute, IN**
- May 2016 Worked with project team to design various products for several clients.
- Jun. 2015 – **Digital Design Intern, Texas Instruments, Dallas, TX**
- Aug. 2015 Performed design verification for high speed I/O IPs.
- Sep. 2013 – **Project Intern, Rose-Hulman Ventures, Terre Haute, IN**
- May 2015 Worked with project team to design various products for several clients.